The Wishverse

A Formal Logical-Transition System over States

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Abstract

We introduce the *Wishverse*, a formal mathematical framework for representing desire-driven state transformation. Stripping away physical interpretation, the Wishverse is defined as a triple $(S, \mathcal{W}, \mathcal{R})$ comprising a state space, a set of wish operators, and a rule system governing valid transitions. Wishes are modeled as non-deterministic functions mapping states to sets of successor states, forming a partial monoid under sequential composition. We establish four foundational axioms ensuring closure, logical consistency, transitional coherence, and well-foundedness under self-referential modification. We further develop notions of wish equivalence, impossibility, and meta-wishes, and identify the Wishverse as a full abstract rewriting system with branching operators. Three directions for further development are outlined: algebraic classification, modal logical embedding, and expressiveness analysis.

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1. Introduction

The concept of a wish—an expressed desire for a transformation of the world—has long been treated informally in folklore, philosophy, and speculative fiction. In all such treatments, wishes function as operators: they receive a current state of the world and produce, or attempt to produce, a different one. Yet no rigorous mathematical treatment of wishes as formal objects has been developed.

This paper introduces the *Wishverse*: a purely abstract, physics-free formal system in which wishes are first-class mathematical objects. We define the Wishverse as a triple $(S, \mathcal{W}, \mathcal{R})$ and develop its algebraic and logical properties from minimal assumptions. The resulting structure is a natural instance of an abstract rewriting system enriched with a meta-operator layer.

Motivation. Our motivation is twofold. First, we seek to understand the structural properties of desire-driven transformation independent of any particular physical, computational, or metaphysical substrate. Second, the Wishverse provides a novel abstract system that connects naturally to modal logic, process algebra, and the theory of non-deterministic automata, suggesting productive cross-domain applications.

Organization. Section 2 defines the primitive triple. Sections 3 and 4 define states and wish operators. Section 5 presents the four axioms. Sections 6–8 develop composition, identity, impossibility, and equivalence. Section 9 establishes the algebraic structure. Section 10 introduces meta-wishes. Section 11 outlines future directions. Section 12 concludes.

2. Primitive Structure

Definition 2.1 (Wishverse)**.** A *Wishverse* is a triple

$$\mathcal{W} \;=\; (S,\; \mathcal{W},\; \mathcal{R}),$$

where S is a non-empty set of *states*, $\mathcal{W}$ is a set of *wish operators*, and $\mathcal{R}$ is a *rule system* governing valid transitions.

No physics, no semantics, and no interpretation are assumed beyond this triple. The system is a purely formal logical-transition structure. The rule system $\mathcal{R}$ is operationalized through the four axioms of Section 5.

3. States

A state $s \in S$ is a complete description of a world configuration. We impose only the following minimal assumptions.

Axiom R1 (State Axioms)**.** (i) Equality is defined: $s = s'$ is a well-formed relation on S.

(ii) States can differ: there exist $s, s' \in S$ with $s \neq s'$.

(iii) $S \neq \emptyset$.

No internal structure is assumed for states at this level of abstraction. The content of a state—whether propositional, relational, or otherwise—is left to interpretation, making the framework maximally general.

4. Wishes as Operators

Definition 4.1 (Wish Operator)**.** A *wish* is a function

$$w : S \longrightarrow \mathcal{P}(S),$$

where $\mathcal{P}(S)$ denotes the power set of S. The set of all wishes is $\mathcal{W}$.

This definition captures the essential non-determinism of wishes: a wish applied to a state does not necessarily yield a unique outcome, but rather a set of possible successor states permitted under the rule system $\mathcal{R}$. Wishes are therefore *non-deterministic transition rules.*

Definition 4.2 (Reachability)**.** The *one-step reachability* relation $\vdash \subseteq S \times S$ is defined by

$$s \vdash s' \quad \Longleftrightarrow \quad \exists\, w \in \mathcal{W} : s' \in w(s).$$

Its reflexive-transitive closure is denoted $\vdash^*$.

5. Axioms of the Rule System

The rule system $\mathcal{R}$ is governed by four axioms.

Axiom R2 (Closure)**.** For every $w \in \mathcal{W}$ and $s \in S$,

$$w(s) \subseteq S.$$

Axiom R1 is a necessary sanity condition: wishes cannot produce objects outside the state space. It holds automatically when wishes are defined as functions into $\mathcal{P}(S)$ as in Definition 4.1.

Axiom R3 (Non-Contradiction of States)**.** States are internally consistent. For any proposition P and any $s \in S$,

$$P \in s \implies \neg P \notin s.$$

Axiom R4 (Consistency of Transitions)**.** Wishes cannot produce logically inconsistent successor states: for all $w \in \mathcal{W}$, $s \in S$, and $s' \in w(s)$, the state s' is internally consistent in the sense of Axiom R2.

Together, Axioms R2 and R3 ensure that neither existing nor generated states harbor contradictions. Wishes whose images would be inconsistent are not ill-formed; they simply map to the empty set (cf. Definition 7.3).

Axiom R5 (Well-Foundedness of Meta-Wishes)**.** The meta-wish modification order (Section 10) is well-founded. Equivalently, there exists no infinite strictly descending chain

$$w_1 \succ w_2 \succ w_3 \succ \cdots$$

under the modification relation.

Axiom R4 prevents collapse into pathological self-reference and ensures tractability of the meta-level. Its role is analogous to the axiom of regularity in set theory.

6. Composition of Wishes

Definition 6.1 (Sequential Composition)**.** Given $w_1, w_2 \in \mathcal{W}$, their *sequential composition* $w_2 \circ w_1 \in \mathcal{W}$ is defined by

$$(w_2 \circ w_1)(s) = \bigcup_{s' \in w_1(s)} w_2(s'), \qquad s \in S.$$

Semantically, w_1 expands the current state into a set of possible successors, and w_2 acts on each, taking their union. This makes $\mathcal{W}$ into a transition algebra.

Proposition 6.2 (Closure under Composition)**.** *If $w_1, w_2 \in \mathcal{W}$, then $w_2 \circ w_1 \in \mathcal{W}$.*

Proof. Let $s \in S$. By Axiom R1 applied to w_1, every $s' \in w_1(s)$ lies in S. By Axiom R1 applied to w_2, each $w_2(s') \subseteq S$. Therefore

$$(w_2 \circ w_1)(s) = \bigcup_{s' \in w_1(s)} w_2(s') \subseteq S,$$

so $w_2 \circ w_1$ satisfies Axiom R1 and is a valid wish. □

Proposition 6.3 (Associativity)**.** *For all $w_1, w_2, w_3 \in \mathcal{W}$,*

$$(w_3 \circ w_2) \circ w_1 = w_3 \circ (w_2 \circ w_1).$$

Proof. Let $s \in S$. Computing the left-hand side:

$$((w_3 \circ w_2) \circ w_1)(s) = \bigcup_{s' \in w_1(s)} (w_3 \circ w_2)(s') = \bigcup_{s' \in w_1(s)} \bigcup_{s'' \in w_2(s')} w_3(s'').$$

Computing the right-hand side:

$$(w_3 \circ (w_2 \circ w_1))(s) = \bigcup_{s'' \in (w_2 \circ w_1)(s)} w_3(s'') = \bigcup_{s'' \in \bigcup_{s' \in w_1(s)} w_2(s')} w_3(s'').$$

The identity

$$\bigcup_{s'' \in \bigcup_i A_i} B_{s''} = \bigcup_i \bigcup_{s'' \in A_i} B_{s''}$$

(which holds for any indexed family of sets) shows that the two expressions coincide. Since s was arbitrary, the equality holds pointwise, hence as functions. □

7. Identity and Impossibility

7.1. The Identity Wish

Definition 7.1 (Identity Wish)**.** The *identity wish* $w_{\mathrm{id}} \in \mathcal{W}$ is defined by

$$w_{\mathrm{id}}(s) = \{s\}, \qquad s \in S.$$

Proposition 7.2 (Identity Law)**.** *For all $w \in \mathcal{W}$,*

$$w \circ w_{\mathrm{id}} = w \qquad \textit{and} \qquad w_{\mathrm{id}} \circ w = w.$$

Proof. Let $s \in S$.

Right identity.

$$(w \circ w_{\mathrm{id}})(s) = \bigcup_{s' \in w_{\mathrm{id}}(s)} w(s') = \bigcup_{s' \in \{s\}} w(s') = w(s).$$

Left identity.

$$(w_{\mathrm{id}} \circ w)(s) = \bigcup_{s' \in w(s)} w_{\mathrm{id}}(s') = \bigcup_{s' \in w(s)} \{s'\} = w(s).$$

In both cases the result equals $w(s)$; since s was arbitrary, the equalities hold as functions. □

7.2. Impossible Wishes

Definition 7.3 (Impossibility)**.** A wish $w \in \mathcal{W}$ is *impossible at* $s \in S$ if $w(s) = \emptyset$. It is *globally impossible* if $w(s) = \emptyset$ for all $s \in S$.

Remark 7.4. Impossibility is a first-class formal object in the Wishverse. When a wish maps to the empty set, no valid successor state exists; the wish fails and the system has no permitted continuation from that state under that wish.

Proposition 7.5 (Absorption of Impossibility)**.** *Let $w_1(s) = \emptyset$ for some $s \in S$. Then for any $w_2 \in \mathcal{W}$,*

$$(w_2 \circ w_1)(s) = \emptyset.$$

Proof.

$$(w_2 \circ w_1)(s) = \bigcup_{s' \in w_1(s)} w_2(s') = \bigcup_{s' \in \emptyset} w_2(s') = \emptyset. \qquad \square$$

Proposition 7.5 confirms that impossibility propagates forward under composition: once a wish fails at a given state, no subsequent wish can recover a successor from that point.

8. Wish Equivalence and the Quotient Space

Definition 8.1 (Wish Equivalence)**.** Two wishes $w_1, w_2 \in \mathcal{W}$ are *equivalent*, written $w_1 \equiv w_2$, if

$$w_1(s) = w_2(s) \qquad \text{for all } s \in S.$$

Proposition 8.2 ($\equiv$ is an Equivalence Relation)**.** *The relation $\equiv$ is reflexive, symmetric, and transitive on $\mathcal{W}$.*

Proof. Reflexivity. For any $w \in \mathcal{W}$ and $s \in S$, $w(s) = w(s)$. Hence $w \equiv w$.

Symmetry. Suppose $w_1 \equiv w_2$. Then $w_1(s) = w_2(s)$ for all s. Since set equality is symmetric, $w_2(s) = w_1(s)$ for all s, so $w_2 \equiv w_1$.

Transitivity. Suppose $w_1 \equiv w_2$ and $w_2 \equiv w_3$. Then for all s, $w_1(s) = w_2(s)$ and $w_2(s) = w_3(s)$, so $w_1(s) = w_3(s)$. Hence $w_1 \equiv w_3$. $\square$

Proposition 8.3 (Congruence with Composition)**.** *If $w_1 \equiv w_1'$ and $w_2 \equiv w_2'$, then $w_2 \circ w_1 \equiv w_2' \circ w_1'$.*

Proof. Let $s \in S$. Since $w_1 \equiv w_1'$, we have $w_1(s) = w_1'(s)$. Since $w_2 \equiv w_2'$, for each $s' \in S$ we have $w_2(s') = w_2'(s')$. Therefore,

$$(w_2 \circ w_1)(s) = \bigcup_{s' \in w_1(s)} w_2(s') = \bigcup_{s' \in w_1'(s)} w_2'(s') = (w_2' \circ w_1')(s). \qquad \square$$

Definition 8.4 (Quotient Wish Space)**.** The *quotient wish space* is $\mathcal{W}/\equiv$, whose elements are the equivalence classes $[w]_\equiv$. Composition descends to $\mathcal{W}/\equiv$ via $[w_2] \circ [w_1] := [w_2 \circ w_1]$, which is well-defined by Proposition 8.3.

9. Algebraic Structure

Proposition 9.1 (Partial Monoid)**.** $(\mathcal{W}, \circ, w_{\mathrm{id}})$ *is a* partial monoid *of non-deterministic state transformers.*

Proof. We verify the three conditions.

Closure. For $w_1, w_2 \in \mathcal{W}$, $w_2 \circ w_1 \in \mathcal{W}$ by Proposition 6.2.

Associativity. Established in Proposition 6.3.

Identity. w_{id} is a two-sided identity by Proposition 7.2.

Partiality. When $w_1(s) = \emptyset$, composition produces $\emptyset$ at s (Proposition 7.5), so the monoid is partial rather than total. □

Theorem 9.2 (Wishverse as an Abstract Rewriting System)**.** *The Wishverse* $\mathcal{W} = (S, \mathcal{W}, \mathcal{R})$ *is a full abstract rewriting system with branching operators.*

Proof. An abstract rewriting system (ARS) is a pair $(A, \to)$ where A is a set and $\to \subseteq A \times A$ is a binary relation [1]. We verify that the Wishverse satisfies, and extends, this structure.

Carrier. Take $A = S$.

Labeled transitions. For each $w \in \mathcal{W}$, define $\to_w \subseteq S \times S$ by $s \to_w s'$ iff $s' \in w(s)$. This gives a family of labeled, non-deterministic transition relations.

Branching. Since $w : S \to \mathcal{P}(S)$, each $\to_w$ may be multi-valued, yielding branching transitions.

Composition and identity. Definition 6.1 provides a composition law on $\mathcal{W}$, and Definition 7.1 provides an identity, making $(\mathcal{W}, \circ, w_{\mathrm{id}})$ a partial monoid (Proposition 9.1).

Impossibility. The empty-set image (Definition 7.3) provides a distinguished absorbing element.

Axioms. Axiom R1 ensures transitions remain in S; Axioms R2 and R3 ensure logical consistency of all states; Axiom R4 ensures well-foundedness of the meta-level.

Together, these components constitute the full structure of an ARS enriched with branching operators, a composition monoid on labels, and a meta-level. □

Corollary 9.3. *The quotient* $(\mathcal{W}/\equiv, \circ, [w_{\mathrm{id}}])$ *is a partial monoid, and the Wishverse descends to an ARS on equivalence classes of wish types.*

Proof. Well-definedness of composition on $\mathcal{W}/\equiv$ follows from Proposition 8.3. Associativity, identity, and partiality descend directly from $(\mathcal{W}, \circ, w_{\mathrm{id}})$. □

10. Meta-Wishes: Self-Reference

Definition 10.1 (Meta-Wish)**.** A *meta-wish* is a function $\mu : \mathcal{W} \to \mathcal{W}$.

Meta-wishes introduce a second-order layer into the Wishverse. A meta-wish can modify the behavior of existing wishes, disable them, or generate new ones. This is the formal analogue of a wish *about* wishes—for example, a wish to make all future wishes more powerful, or to constrain the wishes of others.

Definition 10.2 (Meta-Wish Modification Order)**.** Write $\mu_1 \prec \mu_2$ if μ_2 modifies μ_1, i.e., there exists $w \in \mathcal{W}$ such that $\mu_2(w) \neq \mu_1(w)$ in a manner that depends on μ_1.

Proposition 10.3 (Well-Foundedness of Meta-Wish Chains)**.** *Every strictly descending chain $\mu_1 \succ \mu_2 \succ \mu_3 \succ \cdots$ under $\prec$ is finite.*

Proof. By Axiom R4, the modification order $\prec$ on meta-wishes is well-founded. A partial order is well-founded if and only if it contains no infinite strictly descending chain. Axiom R4 asserts precisely this condition for $\prec$. Hence every descending chain terminates in finitely many steps. □

Remark 10.4. Axiom R4 is not derivable from Axioms R1–R3; it must be postulated independently. Without it, one could construct a meta-wish μ satisfying $\mu(\mu) = \mu'$ for a different meta-wish μ', leading to an infinite regress. Axiom R4 is the direct analogue of the axiom of foundation in ZF set theory.

11. Directions for Further Development

The Wishverse as developed here is already a rich formal structure. We identify three principal directions for deeper investigation.

11.1. Algebraic Classification

The most immediate algebraic question concerns the internal classification of wish types. One may ask: which subsets of $\mathcal{W}/\equiv$ form submonoids of $(\mathcal{W}, \circ)$? Which wish types are invertible, and what is the structure of the invertible fragment? Can one define *homomorphisms* between Wishverses—structure-preserving maps translating one wish system into another?

Such homomorphisms would allow comparison of Wishverses of different expressiveness and identification of universal Wishverses that simulate all others. In the language of category theory [4], Wishverses and their homomorphisms would form a category, and one could study limits, colimits, and adjunctions within this setting.

11.2. Modal Logic Embedding

The Wishverse maps naturally onto a modal logic framework. States carry propositional content (abstractly, via Axiom R2), and wishes act as modal operators: given a wish w and a proposition P, define

$$[w]\,P \;\equiv\; \text{“all successor states under } w \text{ satisfy } P\text{”}, \qquad \langle w\rangle\,P \;\equiv\; \text{“some successor state under } w \text{ satisfies } P\text{”}.$$

This yields a *Wish Logic*: a normal modal logic parameterized by the wish algebra $(\mathcal{W}, \circ)$. The identity wish corresponds to the trivial modality; impossible wishes correspond to inconsistent modalities; and meta-wishes correspond to second-order modal operators. Axioms R1–R4 translate into frame conditions on the corresponding Kripke semantics [2]. This embedding connects the Wishverse to dynamic logic [5] and process algebra [3].

11.3. Expressiveness and Completeness

A third direction concerns the expressiveness of wish systems. Given a Wishverse $\mathcal{W}$ and an initial state s_0, define the *reach* of $\mathcal{W}$ from s_0 as

$$\mathrm{Reach}(\mathcal{W}, s_0) \;=\; \{\, s \in S : s_0 \vdash^* s \,\}.$$

One may then ask: when is $\mathcal{W}$ *complete*, in the sense that $\mathrm{Reach}(\mathcal{W}, s) = S$ for every $s \in S$? When are two Wishverses equivalent in expressiveness? Is there a wish system of minimal cardinality generating a given reach? These questions connect the Wishverse to the theory of generators and relations in group theory, to reachability analysis in concurrent systems [3], and to the theory of term rewriting [1].

12. Conclusion

We have defined the Wishverse as a formal triple $(S, \mathcal{W}, \mathcal{R})$, established four foundational axioms (Closure, Non-Contradiction of States, Consistency of Transitions, and Well-Foundedness of Meta-Wishes), developed the algebraic structure of the wish space as a partial monoid of non-deterministic state transformers, introduced wish equivalence and the quotient wish space, formalized impossibility as a first-class mathematical object with absorbing propagation, and extended the framework to meta-wishes governed by a well-foundedness condition.

The Wishverse is, in its final form, a full abstract rewriting system with branching operators (Theorem 9.2). It provides a novel formal setting that connects naturally to modal logic, process algebra, and reachability theory. The three directions identified—algebraic classification, modal embedding, and expressiveness analysis—offer substantial opportunities for rigorous mathematical development.

The formalization of wishes as mathematical objects opens a productive domain at the intersection of formal methods, abstract algebra, and philosophical logic. We offer the Wishverse as a foundation for this investigation.

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